



Parkinson's Disease: are some cases in fact misdiagnosed Cranio-Mandibular Dysfunction?

Dwight Jennings DDS, MICCMO
Alameda, California

Abstract: Multiple case histories have shown that neuromuscular orthopedic repositioning of the mandible can resolve symptoms normally attributable to Parkinson's disease in some instances. The purpose of this article is to 1) present case histories that mimic Parkinson's 2) describe common neurological disturbances 3) and describe differential diagnostic tests to identify cranio-mandibular dysfunction. **Conclusion:** a review of medical literature and clinical experience indicates that cranio-mandibular dysfunction can induce symptoms normally attributable to Parkinson's disease. The frequency of misdiagnosis in Parkinson's is not known and needs further investigation.

Parkinson's is a diagnosis of exclusion.¹ Emerging clinical evidence has shown that some patients with symptoms normally attributed to Parkinson's are in fact neuropathic manifestations induced by cranio-mandibular dysfunction.

The purpose of this article is to 1) present case histories that mimic Parkinson's 2) describe common neurological disturbances 3) and describe differential diagnostic tests to identify cranio-mandibular dysfunction.

Case histories from multiple dentists has shown that neuromuscular orthopedic repositioning of the cranio-mandibular relationship² is effective in some cases at stopping many symptoms typically attributed to Parkinson's disease Those symptoms include:

- **tremors**
- **limb weakness**
- **rigidity**
- **nocturia**
- **speech disturbance**
- **swallow disturbance**
- **fatigue**
- **depression**
- **anxiety**
- **memory loss**
- **insomnia**
- **dizziness**
- **fainting**
- **impaired proprioception**
- **loss of smell**
- **dystonia**
- **imbalance**
- **gait dysfunction**



- **scoliosis**

Case History:

GD: Age 68. Twenty year working diagnosis of Parkinson's disease with progressive development of severe dystonia. Additional medical complaints of sleep apnea, frozen right shoulder, loss of motor control of right arm except wrist twisting, severe cervicalgia, weakness, and weight loss. Neuromuscular computerized jaw motion analysis showed him to have a severe excess freeway³ space of 7mm. Mandibular orthosis was delivered to support relaxed neuromuscular resting position of the mandible with 1mm freeway space. Patient had an immediate near complete cessation of tremors and dystonia. Monthly adjustment of orthosis has maintained tremor free status. Arm and shoulder function gradually returned. Increase in weight started after four months of maintenance on dental orthosis.

Cranio-mandibular Dysfunction and its Sequella

Cranio-mandibular dysfunction is poorly understood by the dental profession, and even less by the medical profession. It is not the purpose of this article to discuss the many misunderstood aspects of this malady. Suffice it to say, it is my opinion that the medical and dental professions have failed miserably in their knowledge and understanding of the breath of pathology induced by cranio-mandibular dysfunction. This I believe has occurred due to dentistry's lack of knowledge and training in cranio-mandibular orthopedic neurophysiology.

Since most health practitioners are unaware of the means by which cranio-mandibular dysfunction can cause many of the symptoms attributed to Parkinson's disease, I would like to detail a few of those pathways:

Rigidity:

Trigeminal proprioceptors (alignment sensors in jaw muscles) are known to modulate the golgi tendon apparatus systemically (i.e. when the bite is off, the brain knows it and can cause muscles systemically to tighten up). This has been extensively investigated in Japan, typically showing that when the bite is compromised (loss of height on posterior teeth) on laboratory animals on one side of their body, the other side contracts and becomes rigid, thus causing scoliosis. In these studies, the animals become weak on the same side as the scoliosis and quit using their limbs on that side.

Weakness:

Muscle weakness is present in early signs of Parkinson's disease. The work of John Beck, MD, orthopedic surgeon and sport specialist, as well as others⁴, has demonstrated the profound effect that the bite can have on systemic muscle strength. It is his opinion that when the bite is compromised it overloads the autonomic nervous system, causing the body to shutdown (i.e. weaken).⁵

**Depression/Anxiety:**

There is extensive literature associating TMJ (temporomandibular joint dysfunction) with depression. This is thought to be caused by multiple pathways including trigeminal nerve modulation of sensory input into the limbic brain, and pain neuropeptide modulation of brain neurotransmitters.⁶

Fatigue:

Fatigue is a symptom that is frequently reported with cranio-mandibular dysfunction.⁷ Multiple dentists have reported a very high degree of success in reversing chronic fatigue symptoms with bite therapy.

Dizziness/fainting:

The trigeminal nerve is known to control blood flow to the brain through what is called the trigeminal-vascular complex.⁸ Excess trigeminal stimulation has been demonstrated to cause dizziness and fainting.⁹

Gait/impaired proprioception:

With bite disturbance, substance P levels are disturbed, which are known to be associated with disturbed proprioception.¹⁰ The trigeminal nerve modulates somatosensory input into the brain; hence tonicities within the nerve could affect motor function.¹¹

Loss of smell:

Few medical personnel are aware that we have two noses: the olfactory and the trigeminal chemoreceptors. Research has shown that these two systems are interdependent. Hence, with trigeminal disturbance from bite dysfunction, it is easy to see how smell could be affected.¹²

Imbalance:

The vestibular system receives input from a wide variety of sources. The trigeminal nerve, with its massive sensory component (28% of sensory cortex), is a major contributor to vestibular function. Hence, when trigeminal nerve tonicity is disturbed, it has the potential to cause imbalance.¹³

Insomnia:

The trigeminal nerve tonicity level becomes elevated with cranio-mandibular dysfunction. The trigeminal nerve has a major influence on the reticular formation in the brain stem, which controls the activation level of the brain. The trigeminal proprioceptors in the mesencephalic nucleus are known to modulate the pineal gland, hence effecting sleep/wake cycles. Also, trigeminal disturbance has a significant effect on serotonin levels, which effects sleep.¹⁴

Tremor:

Multiple case histories have shown an unclear connection between bite and tremors. The suspected mechanism is thought to be trigeminal input into the reticular formation. Brendan Stack, DDS, MS produced a video entitled "Complex TMJ Disorders and Tremors" a couple years ago, video documenting numerous tremor cases that responded to jaw repositioning therapy.

**Weak Voice:**

The recurrent laryngeal nerve, a branch of the vagus nerve, supplies motor function to the voice box (larynx). The trigeminal nerve is known to modulate vagus nerve function under certain situations; hence bite disturbance would have the ability to modulate voice.

Masked Face:

The trigeminal nerve is known to have the ability to modulate muscles of facial expression.

¹⁵ This I have seen demonstrated numerous times clinically when chronic Bell's palsy symptoms resolve with jaw orthopedic therapy.

Scoliosis:

Scoliosis is known to occur more frequently with Parkinson's disease than with the general public.¹⁶ Scoliosis is also known to be caused by bite disturbance.¹⁷

Identifying Cranio-mandibular Dysfunction

When the jaw is malaligned, the trigeminal nerve is hyper activated through the necessary accommodations that it has to perform. This affects all aspects of the trigeminal nerve: sensory, motor, and proprioceptive. Hence, any and all aspects of trigeminal neural interaction can be affected, leading to a very broad array of disorders. Many studies to date have shown that TMJ patients have a very high medical utilization rate (i.e. they are sick for a very large number of other reasons).¹⁸ If a patient has significant cranio-mandibular dysfunction, a thorough medical history will typically reveal multiple pathologies attributable to trigeminal disturbance. This knowledge is a major factor in the recognition of cranio-mandibular dysfunction.

A major reason for the consequential medical disorders is due to the fact that jaw malalignment causes "substance P" (SP)¹⁹ levels to become elevated in the body. The trigeminal nerve has an extreme high density of pain fibers, thus causing it to have a major effect on SP levels in the body. SP is not actually a neurotransmitter, but a neuropeptide (a compound that does not get recycled, hence it lingers in the body and with endocrine like properties). There is a classification of medical disorders called "neurogenic inflammatory disorders" that are associated with elevated SP. Clinical experience has shown that these disorders respond at a very high percentage rate to orthopedic jaw alignment therapy. This group includes headaches, fibromyalgia, asthma, seizures, etc. SP levels have been implicated in the development of Parkinson's disease.²⁰

Diagnosis of cranio-mandibular dysfunction is quite controversial in the dental profession and quite unreliably performed. Bernard Jankelson placed jaw function on an objective scientific basis with the development of neuromuscular concepts²¹ and electronic instrumentation in the 1970's. These concepts were a great advancement, though they have received limited acceptance by the dental profession.



When screening for cranio-mandibular dysfunction, I have found a couple simple tests of inordinate value. They are:

1. compare position of the mandible in speech and biting: speech and centric occlusion should be on the same trajectory²²
2. have the patient open wide and close on a tongue blade. There should exist a fair arc in open/close without anterior/posterior shifting of the mandible. If the mandible anteriorizes more than 1/8 inch when closing on the tongue blade, it suggests there is a significant cranio-mandibular dysfunction.²³
3. observe the mandibular position of patient at various times: rest position and speech position. The mandible should be close vertically to bite position.²⁴ Excess space between jaws at rests can cause slight speech distortion (lisp). This is often either caused by mouth breathing, particularly nocturnally, or tongue thrusting.

Satisfying these conditions infers the following corollaries:

- a. speech and centric occlusion should be on the same trajectory
- b. there should exist a fair arc in open/close without anterior/posterior shifting of the mandible
- c. Overbites are generally dysfunctional.

Treatment for Cranio-mandibular Dysfunction

When diagnostics²⁵ have confirmed that cranio-mandibular dysfunction is present, treatment is initiated with a diagnostic orthosis to reposition the mandible. This is typically a removable dental appliance that supports the mandible in a relaxed neuromuscular position. This effectuates a calming of the trigeminal system and in time typically resolves many associated pathologies.

In the event that the diagnostic orthosis is effective, the mandible will need to be stabilized in the therapeutic position by a variety of options.

Conclusion:

Both clinical experience and medical literature support the fact that some patients diagnosed with Parkinson's disease, may in fact have neuropathies induced by cranio-mandibular dysfunction.



Dwight Jennings, DDS, MICCMO
2187 Harbor Bay Parkway
Alameda, Ca 94502
510.522.6828
dejdds@aol.com

Definitions:

Cranio-mandibular malalignment- a condition in which the teeth do not relate the maxillary and mandibular bones in the same relationship as where the muscles suspending the jaw want the jaw to be. This requires the jaw muscles to accommodate to the tooth position, thus causing the jaw muscles and trigeminal nerve to become hypertonic/hyperactive. This is a very common condition in modern man.

Trigeminal nerve: the fifth cranial nerve composed of three divisions which exit the skull above the eye, below the eye, and inside the lower jaw. It is unusual in that it has both motor and sensory capabilities. Due to its extreme high density of pain fibers, 28% of the sensory cortex is devoted to it alone. It is anatomically and functionally a spinal nerve. Sitting atop the spinal chord it has the ability to modulate the ascending spinal signal, thus modulating spinal input into the brain.

Trigeminal System: the trigeminal nerve and all systems directly modulated. The systems directly modulated include spinal muscle tone, limbic brain, pineal gland, glossopharyngeal nerve, C1-C3, vagus nerve, cerebral blood flow, thyroid function, sympathetic tonicity, gait, muscle strength, peripheral blood flow, neuropeptide levels, hearing, smell, taste, pigmentation, etc.

¹ Aust Fam Physician. 1992 Oct;21(10):1411-6.

Is it Parkinson's disease?

Hely M.

Westmead Hospital, Sydney.

² Clin Oral Implants Res. 2000 Dec;11(6):566-71.



Clinical outcomes of three Parkinson's disease patients treated with mandibular implant overdentures.

Heckmann SM, Heckmann JG, Weber HP.

School of Dental Medicine, University of Erlangen-Nuremberg, Glückstr. 11, 91054 Erlangen, Germany.

³<http://www.dentalglossary.net/definition/>

distance between the upper and lower teeth with the lower jaw in rest position. Normal range should be approximately 1 to 2 millimeters.

⁴ Clinical Management of Head, Neck and TMJ Pain and Dysfunction. ed. by Harold Gelb. Philadelphia, wb Saunders Co. 1985

Cranio. 2004 Oct;22(4):268-75.

Affecting upper extremity strength by changing maxillo-mandibular vertical dimension in deep bite subjects.

Abdallah EF, Mehta NR, Forgione AG, Clark RE.

Craniofacial Pain Center, 6th Floor, Tufts University School of Dental Medicine, 1 Kneeland St. Boston, MA 02111, USA. emad.abdallah@tufts.edu

Cranio. 2002 Oct;20(4):264-73.

Comment in:

Cranio. 2003 Apr;21(2):86.

The effect of stepwise increases in vertical dimension of occlusion on isometric strength of cervical flexors and deltoid muscles in nonsymptomatic females.

Chakfa AM, Mehta NR, Forgione AG, Al-Badawi EA, Lobo SL, Zawawi KH.

Cranio. 1999 Apr;17(2):85-92.

Comment in:

Cranio. 2002 Apr;20(2):79.

The effect of vertical dimension and mandibular position on isometric strength of the cervical flexors.

al-Abbasi H, Mehta NR, Forgione AG, Clark RE.

Implant Center, Tufts University School of Dental Medicine, Massachusetts 02111, USA.

⁵ Beck, J. Personal conversations 2007

Eur Neurol. 1986;25(2):130-3.

Muscle strength testing in Parkinson's disease.



Koller W, Kase S.

⁶ J Orofac Pain. 2007 Winter;21(1):46-54.

Oral health-related quality of life in patients with temporomandibular disorders.

John MT, Reissmann DR, Schierz O, Wassell RW.

Department of Prosthodontics and Materials Science, University of Leipzig, Germany.
mike.john@medizin.uni-leipzig.de

J Prosthet Dent. 2002 Nov;88(5):479-84.

Depression and somatization in patients with temporomandibular disorders.

Yap AU, Tan KB, Chua EK, Tan HH.

Faculty of Dentistry, Department of Restorative Dentistry, National University of Singapore, 5 Lower Kent Ridge Road, Singapore 119074, Republic of Singapore.
rsdyapuj@nus.edu.sg

Eur J Pharmacol. 1996 Sep 5;311(1):7-14.

Effects of central administration of tachykinin receptor agonists and antagonists on plus-maze behavior in mice.

Teixeira RM, Santos AR, Ribeiro SJ, Calixto JB, Rae GA, De Lima TC.

Department of Pharmacology, Universidade Federal de Santa Catarina, Florianópolis, Brazil.

⁷ **[A primary research on the concomitant symptoms of temporomandibular joint pain]**

[Article in Chinese]

Jiang T, Li J, Zhang ZK.

Department of Prosthodontics, Peking University School of Stomatology, Beijing 100081, China.
janqi@263.net

⁸ *Journal of Cerebral Blood Flow & Metabolism* (1999) **19**, 115–127; doi:10.1097/00004647-199902000-00001

The Trigeminovascular System in Humans: Pathophysiologic Implications for Primary Headache Syndromes of the Neural Influences on the Cerebral Circulation. Arne May and Peter J Goadsby

University Department of Clinical Neurology, Institute of Neurology, The National Hospital for Neurology and Neurosurgery, Queen Square, London, UK

⁹ Nippon Jibiinkoka Gakkai Kaiho. 1999 Jul;102(7):891-7.

[Effect of adrenaline on vagus nerve reflexes]

[Article in Japanese]



Masaki Y, Furukawa T, Watanabe M, Ichikawa G.

Department of Otorhinolaryngology, Juntendo University School of Medicine, Tokyo.

Anesth Prog. 1996 Spring;43(2):41-51.

The effects of the hypothalamus on hemodynamic changes elicited by vagal nerve stimulation.

Niwa H, Hirota Y, Shibutani T, Sugiyama K, Matsuura H.

Department of Dental Anesthesiology, Osaka University, Japan.

¹⁰ Exp Brain Res. 2001 Mar;137(2):249-53.

Participation of NK1 receptors in nociceptin-induced modulation of rat knee joint mechanosensitivity.

McDougall JJ, Hanesch U, Pawlak M, Schmidt RF.

McCaig Centre for Joint Injury and Arthritis Research, 3330, Hospital Drive NW, University of Calgary, Calgary, AB., T2 N 4N1, Canada. mcdougaj@ucalgary.ca

Neurosci Lett. 1984 Sep 28;51(1):61-5.

Demonstration of substance P immunoreactivity in the nucleus dorsalis of human spinal cord.

Pioro EP, Hughes JT, Cuello AC.

¹¹ Neuropathol Appl Neurobiol. 2003 Oct;29(5):418-33.

Anatomically based guidelines for systematic investigation of the central somatosensory system and their application to a spinocerebellar ataxia type 2 (SCA2) patient.

Rüb U, Schultz C, Del Tredici K, Gierga K, Reifenberger G, de Vos RA, Seifried C, Braak H, Auburger G.

Institute for Clinical Neuroanatomy, Johann Wolfgang Goethe University, Theodor-Stern-Kai 7, D-60590 Frankfurt/Main, Germany.

No To Shinkei. 1996 Sep;48(9):845-50.

[Therapeutic effectiveness of combined microsurgery and radiosurgery in a patient with a huge trigeminal neurinoma]

[Article in Japanese]

Yamasaki T, Nagao S, Kagawa T, Takamura M, Moritake K, Tanaka Y, Kida T, Kobayashi T.

Department of Neurosurgery, Shimane Medical University, Japan.

¹² Cereb Cortex. 2007 Oct;17(10):2268-75. Epub 2006 Dec 5.



Interactions between olfaction and the trigeminal system: what can be learned from olfactory loss.

Frasnelli J, Schuster B, Hummel T.

Cognitive Neuroscience Unit, Montreal Neurological Institute, Montreal, Quebec, Canada.
johannes.frasnelli@mcgill.ca

Laryngoscope. 2006 Aug;116(8):1520-2.

Loss of trigeminal sensitivity reduces olfactory function.

Husner A, Frasnelli J, Welge-Lüssen A, Reiss G, Zahnert T, Hummel T.

Smell & Taste Clinic, Department of Otorhinolaryngology, University of Dresden Medical School ("Technische Universität Dresden"), Fetscherstrasse 74, 01307 Dresden, Germany.

¹³ J Comp Neurol. 1999 Jun 21;409(1):153-68.

Organisation of reciprocal connections between trigeminal and vestibular nuclei in the rat.

Buisseret-Delmas C, Compont C, Delfini C, Buisseret P.

Laboratoire de Neuroanatomie Fonctionnelle des Systèmes Sensorimoteurs, Paris, France.

Neurosci Lett. 1999 Apr 2;264(1-3):133-6.

Primary trigeminal afferents to the vestibular nuclei in the rat: existence of a collateral projection to the vestibulo-cerebellum.

Pinganaud G, Bourcier F, Buisseret-Delmas C, Buisseret P.

Laboratoire de Neuroanatomie Fonctionnelle des Systèmes Sensorimoteurs, Paris, France. pinganaud@ccr.jussieu.fr

Adv Otorhinolaryngol. 2002;59:26-34.

Role of substance P in the peripheral vestibular and auditory system.

Felix H, Oestreicher E, Felix D, Ehrenberger K.

ENT Department, University Hospital Zürich, Switzerland. felix@orl.usz.ch

J Reprod Med. 1996 Jun;41(6):397-402.

Neuroendocrine cell-axonal complexes in the minor vestibular gland.

Warner TF, Tomic S, Chang CK.

Department of Pathology, University of Wisconsin Hospital and Clinics, Madison 53792, USA.

Acta Otolaryngol. 1984 May-Jun;97(5-6):523-8.

Is substance P the neurotransmitter in the vestibular end organs?

Ylikoski J, Päivärinta H, Eränkö L, Mrena I, Lehtosalo J.



¹⁴ Nervenarzt. 2006 Oct;77(10):1251-9; quiz 1260.

[Sleep in neurodegenerative disorders]

[Article in German]

Happe S, Mayer G.

Abteilung für Klinische Neurophysiologie, Klinikum Bremen-Ost, Züricher Strasse 40, 28325, Bremen, Deutschland. Svenja.Happe@klinikum-bremen-ost.de

Sleep Med Rev. 2002 Oct;6(5):341-51.

Comment in:

Sleep Med Rev. 2002 Oct;6(5):353-7; discussion 359.

Sleep Med Rev. 2003 Feb;7(1):101-2; author reply 103-5.

Neurobiological bases for the relation between sleep and depression.

Adrien J.

INSERM U288, CHU Pitié-Salpêtrière, 91 Bd de l'Hôpital, 75013 Paris, France.
adrien@ext.jussieu.fr

J Orofac Pain. 2007 Spring;21(2):107-19.

Comment in:

J Orofac Pain. 2007 Spring;21(2):85-6.

Prevalence and impact of post-traumatic stress disorder symptoms in patients with masticatory muscle or temporomandibular joint pain: differences and similarities.

Bertoli E, de Leeuw R, Schmidt JE, Okeson JP, Carlson CR.

Orofacial Pain Center, College of Dentistry, University of Kentucky, Lexington, KY 40532-0297, USA.

¹⁵ J Comp Neurol. 2001 Jan 1;429(1):80-93.

Trigemino-reticulo-facial and trigemino-reticulo-hypoglossal pathways in the rat.

Zerari-Mailly F, Pinganaud G, Dauvergne C, Buisseret P, Buisseret-Delmas C.

Laboratoire de Neuroanatomie Fonctionnelle des Systèmes Sensorimoteurs, 75251 Paris cedex 05, France. zerari@ccr.jussieu.fr

¹⁶ Duvoisin RC, Marsden CD. Note on the scoliosis of Parkinsonism. *J Neurol Neurosurg Psychiatry* 1975;38:787-93.

MURATA HIDEYUKI(Hamamatsu Univ. School of Medicine, Fac. Medicine, JPN) YAMANASHI AKIHIRO(Hamamatsu Univ. School of Medicine, Fac. Medicine, JPN) NAGANO AKIRA(Hamamatsu Univ. School of Medicine, Fac. Medicine, JPN). The Evaluation of Scoliosis in Patients with Parkinson's Disease. *Journal Title;Spinal Deformity*
ISSN:0911-6826
VOL.20;NO.1;PAGE.26-30(2005)



¹⁷ Sumioka T: systemic effects of the peripheral disturbance of the trigeminal system: influences of the occlusal destruction in dogs. J Kyoto Pref. Univ. Med. 1989, 98, 1077-1085.

¹⁸ Cranio. 1998 Jul;16(3):185-93. Links
Health care utilization by patients with temporomandibular joint disorders.

* Shimshak DG,
* DeFuria MC.

Management Science & Information Department, University of Massachusetts, Boston 02125, USA.

¹⁹ Substance P is the neurotransmitter from C fibers (pain sensors) that sends all pain signals to the brain.

²⁰ Neurosci Lett. 1993 Oct 1;160(2):163-6.

Parallel alterations in Met-enkephalin and substance P levels in medial globus pallidus in Parkinson's disease patients.

de Ceballos ML, Fernandez A, Jenner P, Marsden CD.

Department of Neuropharmacology, Cajal Institute, CSIC, Madrid, Spain.

Brain. 1996 Jun;119 (Pt 3):823-30.

Alterations in peptide levels in Parkinson's disease and incidental Lewy body disease.

Fernandez A, de Ceballos ML, Rose S, Jenner P, Marsden CD.

Neurodegeneration Group, Cajal Institute, CSIC, Madrid, Spain.

J Neural Transm. 1999;106(9-10):943-8.

Increased cerebrospinal fluid levels of substance P in patients with amyotrophic lateral sclerosis. Short communication.

Matsuishi T, Nagamitsu S, Shoji H, Itoh M, Takashima S, Iwaki T, Shida N, Yamashita Y, Sakai T, Kato H.

Department of Pediatrics and Child Health, Kurume University School of Medicine, Fukuoka, Japan.

²¹ <http://medical-dictionary.thefreedictionary.com/Neuromuscular+dentistry>

Definition: **dentistry, neuromuscular**

n a subdiscipline of dentistry concerned with correcting alignment problems at the temporomandibular joint. This branch of dentistry focuses primarily on caring for the muscles, nerves, and other tissue as opposed to teeth and bones.

For further information on the dental controversy see: http://www.myotronics.com.au/Article_006.htm



²² Speech is a muscular phenomenon and biting is a skeletal phenomenon- you like to have the musculoskeletal system coordinated. Typically, when the discrepancy, either vertically or horizontally, becomes greater than 1/8 inch (3mm), significant trigeminal pathology arises.

²³ Even though the TMJ has the ability to translate, the TMJ functions best when function requires it to translate least. Excess range of motion (including translation) hyper excites the trigeminal system.

²⁴ The temporomandibular joint functions best when occlusion is such that it supports jaw functions within a minimal range of motion (i.e. speech, rest, and open/close arcs are superimposed, as well as height of mandible during speech is within 1.5 mm of occlusion). George W. Bush is a good example of a person with excess space between the teeth when speaking.

²⁵ Diagnostics often includes examination, medical history, appropriate radiographs, and neuromuscular instrumentation assessment (a computer-based system for three dimensional jaw tracking, surface electromyography and temporomandibular joint sound study).